

Exercise - Singleton Pattern (Logger.cs)

The screenshot shows the Visual Studio Code editor with the following components:

- Menu Bar:** File, Edit, Selection, View, Go, Run, Terminal, Help.
- Explorer Panel:** Shows the project structure for 'DEEPSKILLING' with files 'Exercise-1-Singleton.csproj', 'Logger.cs', and 'Program.cs'. A 'TIMELINE' view is also visible.
- Editor Panel:** Displays the code for 'Logger.cs'.

The code in 'Logger.cs' implements the Singleton pattern:

```
1  using System;
2
3  namespace SingletonPatternExample
4  {
5      public class Logger
6      {
7          private static Logger instance;
8
9          private Logger()
10         {
11         }
12
13         public static Logger GetInstance()
14         {
15             if (instance == null)
16             {
17                 instance = new Logger();
18             }
19
20             return instance;
21         }
22
23         public void Log(string message)
24         {
25             Console.WriteLine(message);
26         }
27     }
28 }
```

Program.cs

The screenshot shows the Visual Studio Code interface with the following components:

- Menu Bar:** File, Edit, Selection, View, Go, Run, Terminal, Help.
- Toolbar:** Includes icons for Explorer, Search, Source Control, Run and Debug, and Extensions. The top right shows a tab count of 2 and a chat icon.
- EXPLORER Panel:** Displays the project structure under 'DEEPSKILLING'. The files listed are 'Exercise-1-Singleton.cspj', 'Logger.cs', and 'Program.cs'. The 'Program.cs' file is selected and highlighted.
- Code Editor:** Displays the content of 'Program.cs'. The code is as follows:

```
1 using System;
2
3 namespace SingletonPatternExample
4 {
5     class Program
6     {
7         static void Main(string[] args)
8         {
9             Logger logger1 = Logger.GetInstance();
10            Logger logger2 = Logger.GetInstance();
11
12            logger1.Log("Application Started");
13            logger2.Log("User Logged In");
14
15            Console.WriteLine();
16
17
18            if (logger1 == logger2)
19            {
20                Console.WriteLine("Only one Logger object exists.");
21            }
22            else
23            {
24                Console.WriteLine("Multiple Logger objects created.");
25            }
26        }
27    }
28 }
```

Program Output (Terminal)

File Edit Selection View Go Run Terminal Help

Deepskilling

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EXPLORER

DEEPSKILLING

Exercise-1-Singleton.csproj

Logger.cs

Program.cs

TIMELINE

Exercise-1-Singleton.csp C# Logger.cs

C# Program.cs

1 using System;

2

3 namespace SingletonPatternExample

4 {

5 class Program

6 {

7 static void Main(string[] args)

8 {

9 Logger logger1 = Logger.GetInstance();

10 Logger logger2 = Logger.GetInstance();

11

12 logger1.Log("Application Started");

13 logger2.Log("User Logged In");

14

15 Console.WriteLine();

16

17

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PSC:\Users\Student\Deepskilling\Exercise-1-Singleton> dotnet run

Application Started

User Logged In

Only one Logger object exists.

PSC:\Users\Student\Deepskilling\Exercise-1-Singleton>

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